



ATTENTION, EXAM PARTICIPANTS!

ENTRANCE EXAMINATION · GRADE 9

SURNAME		
NAME		
GRADE	9	

INSTRUCTIONS

1. Fill in your **surname and name** on the answer sheet.
2. **40 questions** are given, and you have **80 minutes** to complete them.
3. Choose **only one correct answer** for each question and **shade the circle completely with an ordinary pencil**. A partial mark or a mark outside the circle is not counted.
4. Erase mistakes with a **soft eraser**, without damaging the paper.
5. **Answer the easy questions first** and come back to the difficult ones later.

NOTE! This paper contains the Mathematics and English sections only.

GRADING SYSTEM

MATHEMATICS

Questions 1-20

2.5 points

ENGLISH

Questions 21-40

1.5 points

SAMPLE OF FILLING IN

HOW TO MARK YOUR ANSWER SHEET

ENGLISH (A, B, C, D, E)						
Ⓐ	Ⓑ	Ⓒ	Ⓓ	Ⓔ	●	
○	○	○	○	○	●	correct
○	○	○	○	○	×	wrong
○	○	○	○	○	×	wrong
○	○	○	○	○	×	wrong
●	○	○	○	○	●	correct

GOOD LUCK!

1. Simplify the fraction: $\frac{x^2 - 6x + 9}{x - 3}$

- A) $x + 3$
- B) $x - 3$
- C) $(x - 3)^2$
- D) $x - 9$
- E) $3 - x$

2. Perform the addition: $\frac{1}{x} + \frac{1}{2x}$

- A) $2/(3x)$
- B) $1/(3x)$
- C) $3/(2x)$
- D) $1/(2x^2)$
- E) $2/x$

3. The graph of the function $y = k/x$ passes through the point (2; 6). Find k .

- A) 3
- B) 8
- C) 4
- D) 12
- E) 6

4. Calculate: $\frac{\sqrt{48}}{\sqrt{3}}$

- A) 16
- B) 4
- C) $2\sqrt{3}$
- D) 8
- E) $\sqrt{45}$

5. Calculate: $27^{2/3}$

- A) 18
- B) 3
- C) 81
- D) 6
- E) 9

6. Solve the inequality: $3x - 7 > 5$

- A) $x > 4$
- B) $x < 4$
- C) $x > -4$
- D) $x \geq 4$
- E) $x > 12$

7. Solve the system of inequalities: $x > 2$ and $x \leq 6$. Write the answer as a numerical interval.

- A) (2; 6)
- B) [2; 6]
- C) (2; 6]
- D) [2; 6)
- E) (6; $+\infty$)

8. Solve the equation: $|x - 3| = 5$

- A) $x = 8$
- B) $x = -2$
- C) $x = 8$ and $x = 2$
- D) $x = 8$ and $x = -2$
- E) $x = -8$ and $x = 2$

9. Solve the quadratic equation:

$$x^2 - 7x + 12 = 0$$

- A) $x = 3$ and $x = 4$
- B) $x = -3$ and $x = -4$
- C) $x = 2$ and $x = 6$
- D) $x = 1$ and $x = 12$
- E) $x = -2$ and $x = -6$

10. Find the discriminant of the equation $x^2 + 4x + 5 = 0$ and determine the number of roots.

- A) $D = 4$, two roots
- B) $D = 0$, one root
- C) $D = -4$, no roots
- D) $D = 36$, two roots
- E) $D = -4$, one root

11. Using Vieta's theorem, find the sum and the product of the roots of the equation $x^2 - 5x + 6 = 0$.

- A) sum 5, product 6
- B) sum -5, product 6
- C) sum 5, product -6
- D) sum 6, product 5
- E) sum -5, product -6

12. Find the sum of the interior angles of a convex pentagon.

- A) 360°
- B) 540°
- C) 720°
- D) 450°
- E) 500°

13. The bases of a trapezoid are 8 cm and 14 cm. Find the midline of the trapezoid.

- A) 10 cm
- B) 12 cm
- C) 22 cm
- D) 11 cm
- E) 6 cm

14. The legs of a right triangle are 9 cm and 12 cm. Find the hypotenuse (Pythagorean theorem).

- A) 13 cm
- B) 21 cm
- C) 16 cm
- D) 14 cm
- E) 15 cm

15. Find the distance between the points A(1; 2) and B(4; 6).

- A) $\sqrt{7}$
- B) 5
- C) 7
- D) $\sqrt{5}$
- E) 25

16. The diagonals of a rhombus are 10 cm and 12 cm. Find the area of the rhombus.

- A) 120 cm^2
- B) 30 cm^2
- C) 55 cm^2
- D) 60 cm^2
- E) 44 cm^2

17. Continue the sequence: 2, 5, 11, 23, 47, ...

- A) 94
- B) 95
- C) 96
- D) 93
- E) 99

18. On an island, knights always tell the truth and liars always lie. You meet A and B. A says: "At least one of us is a liar". Who are they?

- A) Both are knights
- B) Both are liars
- C) A is a knight, B is a liar
- D) A is a liar, B is a knight
- E) It cannot be determined

19. A does a job in 12 days and B does it in 18 days. They worked together for 4 days, then A left. In how many days will B finish the remaining work?

- A) 6
- B) 8
- C) 9
- D) 10
- E) 5

20. How many three-digit numbers can be formed from the digits 1, 2, 3, 4, 5 if the digits are not repeated?

- A) 120
- B) 125
- C) 60
- D) 15
- E) 10

Part 1. READING — Read the text and answer questions 21–25.**The Mystery of Loch Ness**

Loch Ness is a large, deep lake in the north of Scotland. For hundreds of years, people have told stories about a huge creature that supposedly lives in its dark waters. The creature is known all over the world as the Loch Ness Monster, or simply "Nessie". The most famous photograph of Nessie was taken in 1934 and appears to show a long neck rising out of the water. For many years, this picture convinced people that the monster was real.

However, in 1994 it was revealed that the famous photograph was a hoax — the "monster" was actually a small model attached to a toy submarine. Since then, scientists have searched the lake many times using modern equipment, but they have never found any real evidence of a large animal. Some visitors are disappointed by this, yet thousands of tourists still travel to Loch Ness every year. Whether or not Nessie exists, the legend has certainly made this quiet Scottish lake famous around the world.

21. What is the text mainly about?

- A) How to travel to Scotland
- B) The legend of a creature in a Scottish lake
- C) The history of photography
- D) How scientists use submarines
- E) The wildlife of Scotland

22. What did the famous 1934 photograph seem to show?

- A) a toy submarine
- B) a group of tourists
- C) a long neck rising from the water
- D) a scientist with equipment
- E) a small model of a lake

23. What was revealed about the photograph in 1994?

- A) It was taken by a scientist
- B) It was actually a hoax
- C) It showed a real monster
- D) It was lost forever
- E) It was drawn by an artist

24. The word "hoax" in the text is closest in meaning to:

- A) a true story
- B) a scientific discovery
- C) a trick to deceive people
- D) an old photograph
- E) a deep lake

25. Which statement best expresses the writer's conclusion?

- A) Scientists have proved that Nessie is real
- B) Nobody visits Loch Ness any more
- C) The legend has made the lake famous, whether or not Nessie is real
- D) The photograph was never explained
- E) Tourists are no longer interested in the lake

Part 2. VOCABULARY — Questions 26–32. Choose the best option.

26. I've had a terrible _____ all morning; I think I need to lie down in a dark room.

- A) stomach
- B) headache
- C) toothache
- D) temperature
- E) cough

27. You should _____ this new phone with the old one before you decide which to buy.

- A) compare
- B) prepare
- C) repair
- D) appear
- E) declare

28. Excuse me, could you tell me the way to the station? — Yes, go straight on and _____ left at the traffic lights.

- A) turn
- B) return
- C) arrive
- D) cross over
- E) drive past

29. Bees are very important because they _____ flowers, which helps plants to produce fruit.

- A) pollute
- B) pollinate
- C) populate
- D) polish
- E) promote

30. "The scientists made an important discovery." The word "discovery" means something that has been:

- A) hidden forever
- B) found out for the first time
- C) thrown away
- D) sold in a shop
- E) broken by accident

31. A _____ is a piece of writing that gives an opinion about a book or film.

- A) recipe
- B) review
- C) reply
- D) receipt
- E) reminder

32. The bridge was so _____ that it took engineers ten years to build it.

- A) ordinary
- B) simple
- C) complex
- D) cheap
- E) tiny

Part 3. GRAMMAR — Questions 33–40. Choose the correct answer.

33. This is the _____ building in the whole city.

- A) tall
- B) taller
- C) tallest
- D) more tall
- E) most tallest

34. I _____ TV when the electricity suddenly went off.

- A) watch
- B) watched
- C) was watching
- D) am watching
- E) have watched

35. If it rains tomorrow, we _____ at home.

- A) stay
- B) stayed
- C) will stay
- D) would stay
- E) have stayed

36. Cars are _____ expensive than bicycles.

- A) much
- B) more
- C) the most
- D) as
- E) many

37. The old castle _____ built more than five hundred years ago.

- A) is
- B) was
- C) has
- D) did
- E) were

38. Look at those dark clouds — I think it _____ rain.

- A) is going to
- B) goes to
- C) will to
- D) going to
- E) is raining to

39. She isn't here now. She _____ have gone home already.

- A) must to
- B) might
- C) should to
- D) can to
- E) need

40. You _____ smoke in the museum; it is strictly forbidden.

- A) must
- B) have to
- C) mustn't
- D) don't have to
- E) needn't